



ECEN 314—Signals and Systems

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Module 20 - Discrete Time Systems

Learning Objectives

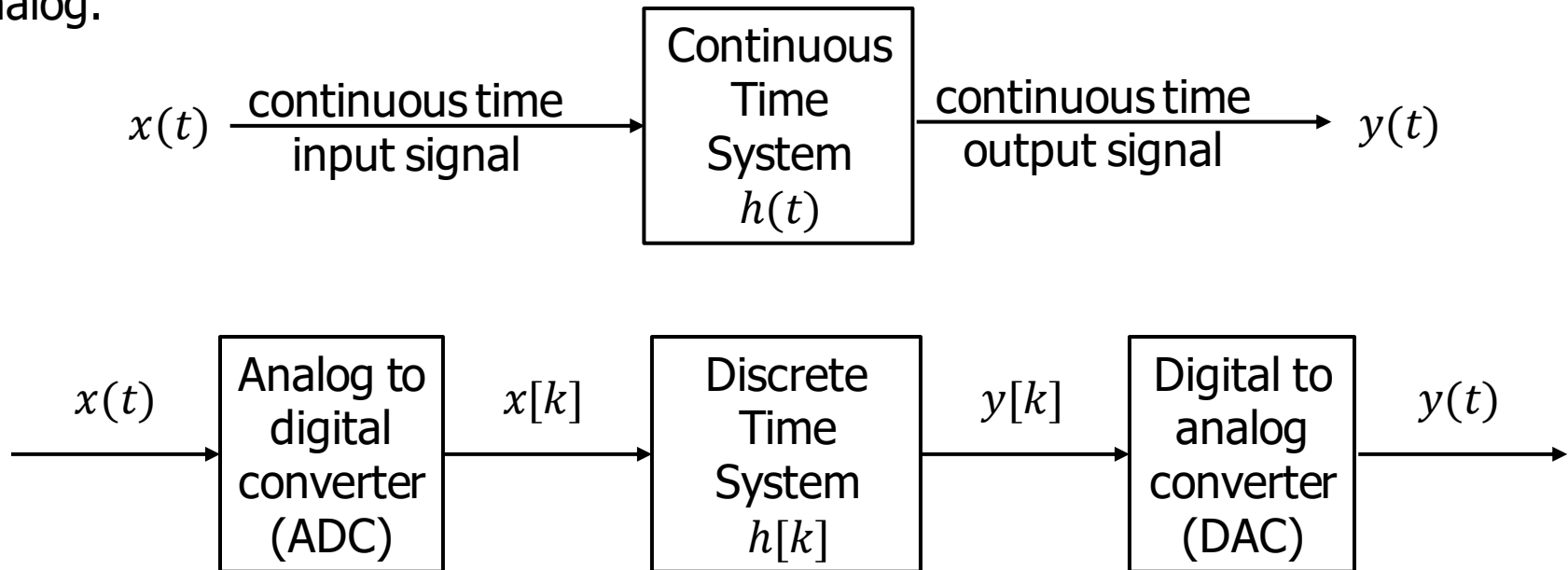


- Understand the different approaches to designing a discrete time system to mimic a continuous time system:
 - Equal input/output relationship
 - Equal impulse response
 - Equal transfer function

Discrete Time Systems



With the prevalence of powerful digital computing capabilities, it is very common to implement a system in discrete time, even if the underlying application is inherently analog.



In this scenario, we would want the equivalent digital system to behave the same way as the original analog system with respect to how it processes the analog input signal.

Even if the ADC and DAC function ideally, how to choose the discrete time system so that the two systems are equivalent can be a bit tricky.

Characterizing Discrete Time Systems



With continuous time systems, we started by describing a system in terms of its input/output relationship. For LTI systems this led to several characterizations:

- ✓ Differential Equation,

$$b_n \frac{d^n y}{dt^n} + \cdots + b_2 \frac{d^2 y}{dt^2} + b_1 \frac{dy}{dt} + b_0 y(t) = a_m \frac{d^m x}{dt^m} + \cdots + a_2 \frac{d^2 x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x(t)$$

- ✓ Impulse response, $h(t) \rightarrow y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(t-v)h(v)dv$.
- ✓ Transfer Function, $H(s)$ or $H(f) \rightarrow Y(s) = H(s)X(s)$ or $Y(f) = H(f)X(f)$.

How do these concepts manifest themselves in discrete time?

- ✓ Differential Equation $\rightarrow$ Difference Equation,

$$\begin{aligned} & b_n y[k-n] + \cdots + b_2 y[k-2] + b_1 y[k-1] + b_0 y[k] \\ & = a_m x[k-m] + \cdots + a_2 x[k-2] + a_1 x[k-1] + a_0 x[k] ?? \end{aligned}$$

- ✓ Impulse response, $h[k]$ ($= h(k\Delta t)??$)
- ✓ Transfer Function, $H[m]$ ($=$ DFT of $h[k]??$)

ODEs in Discrete Time



Let's start by converting a differential equation to discrete time.

$$\frac{dy}{dt} = \lim_{\Delta t \rightarrow 0} \frac{y(t + \Delta t) - y(t)}{\Delta t} \cong \frac{y(t + \Delta t) - y(t)}{\Delta t} = \frac{y[k + 1] - y[k]}{\Delta t}.$$

Suppose we had a simple first order ODE:

$$b_1 \frac{dy}{dt} + b_0 y(t) = a_0 x(t)$$

Applying the discrete approximation to the derivative above leads to a first order difference equation:

$$b_1 \left(\frac{y[k + 1] - y[k]}{\Delta t} \right) + b_0 y[k] = a_0 x[k]$$

$$b_1 y[k + 1] + (b_0 \Delta t - b_1) y[k] = (a_0 \Delta t) x[k]$$

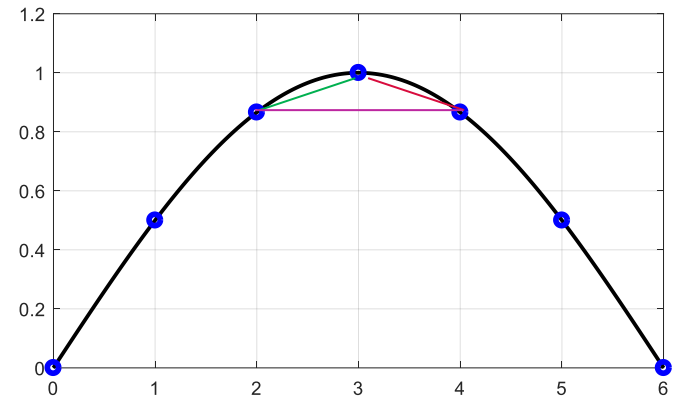
Notice the coefficients of the difference equation are not the same as the coefficients of the underlying differential equation (even if we were to assume that $\Delta t=1$).

ODEs in Discrete Time



Furthermore, there are several ways we might approximate derivatives using discrete differences:

- ✓ Forward Difference: $\frac{dy}{dt} \cong \frac{y[k+1]-y[k]}{\Delta t}$
- ✓ Backward Difference: $\frac{dy}{dt} \cong \frac{y[k]-y[k-1]}{\Delta t}$
- ✓ Central Difference: $\frac{dy}{dt} \cong \frac{y[k+1]-y[k-1]}{2\Delta t}$



All three lead to slightly different difference equations.

Even more ambiguity would arise if we needed to approximate higher order derivatives with finite differences.

- There is no a clear-cut obvious way to convert a continuous time ODE to an equivalent discrete-time difference equation.
- These differences might not be significant if we sampled at a sufficiently high rate (i.e., they all are the same as $\Delta t \rightarrow 0$), but in practice, we often try to get away with the lowest sampling rate that we can.

Impulse Invariance



One way that we could design a DT system to mimic the behavior of a CT system is to choose the impulse response of the DT system to match the samples of the impulse response of the CT system:

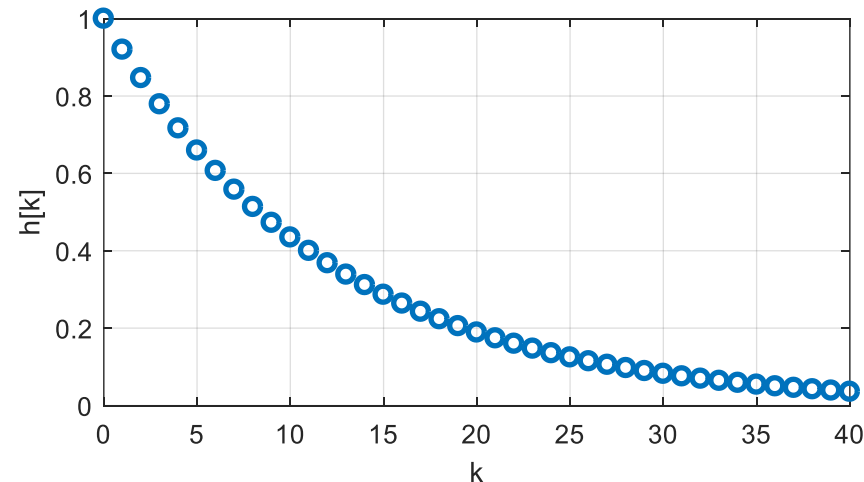
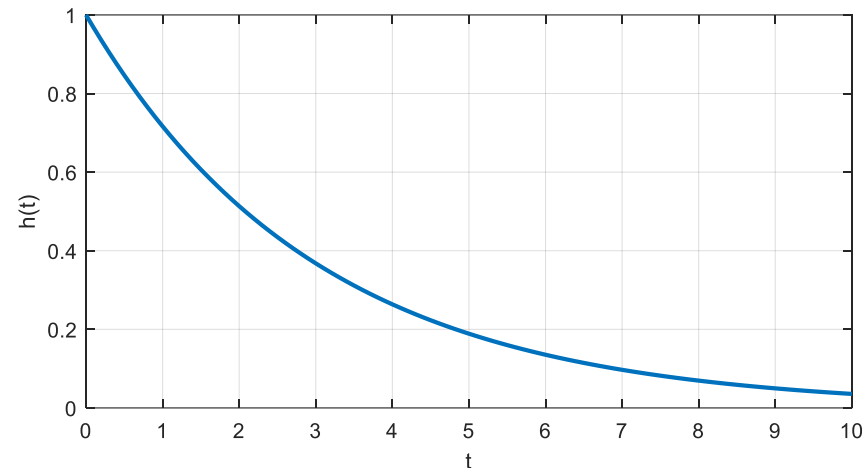
$$h[k] = h(kT_s)$$

The two systems are said to be *impulse invariant* in that the responses of both systems to an impulse at the input are the same.

Example: $h(t) = e^{-at}u(t)$

$$\begin{aligned} h[k] &= h(kT_s) = e^{-akT_s}u(kT_s) \\ &= b^k u[k] \quad \text{where } b = e^{-aT_s} \end{aligned}$$

We will find that the responses of the two systems to other inputs may not necessarily be the same



Impulse Invariance



Consider the example on the previous slide where $h(t) = e^{-at}u(t)$ and $h[k] = b^k u[k]$ with $b = e^{-aT_s}$

Suppose we find the response of each system to a unit step function:

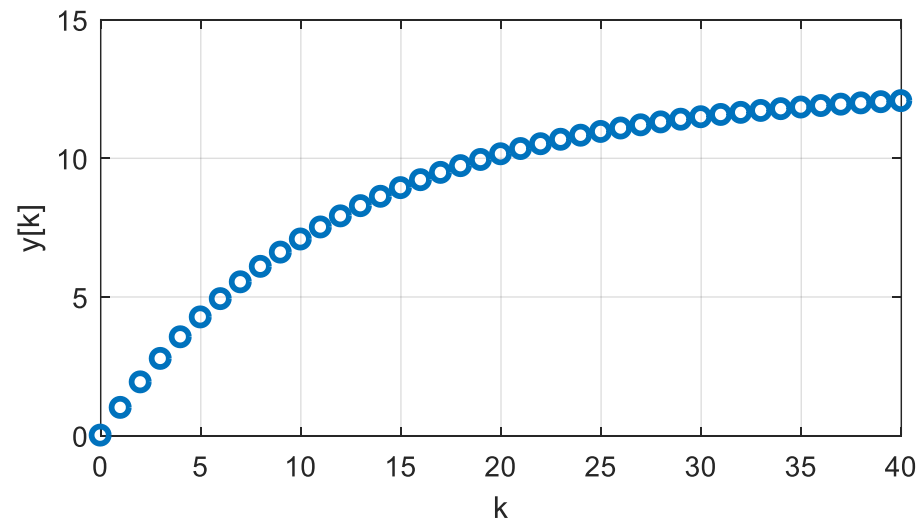
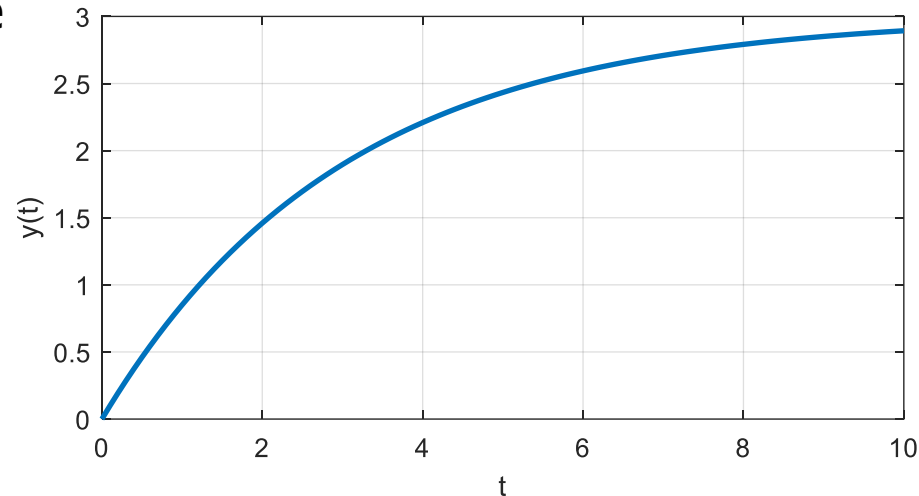
$$y(t) = h(t) * u(t) = \frac{1}{a}(1 - e^{-at})u(t)$$

$$y[k] = h[k] * u[k] = \frac{1 - b^{k+1}}{1 - b}$$

- ✓ The shapes look similar.
- ✓ The limiting values are very different

$$\lim_{t \rightarrow \infty} y(t) = 1/a$$

$$\lim_{k \rightarrow \infty} y[k] = \frac{1}{1 - b} = \frac{1}{1 - e^{-aT_s}}$$



Impulse Invariance



For the previous example, how do the transfer functions compare?

$$H(f) = \frac{1}{a + j2\pi f}$$

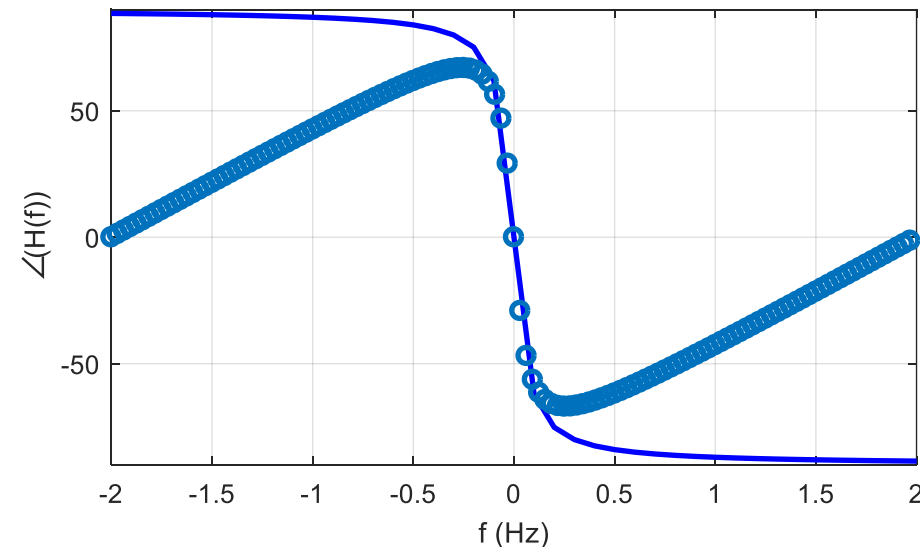
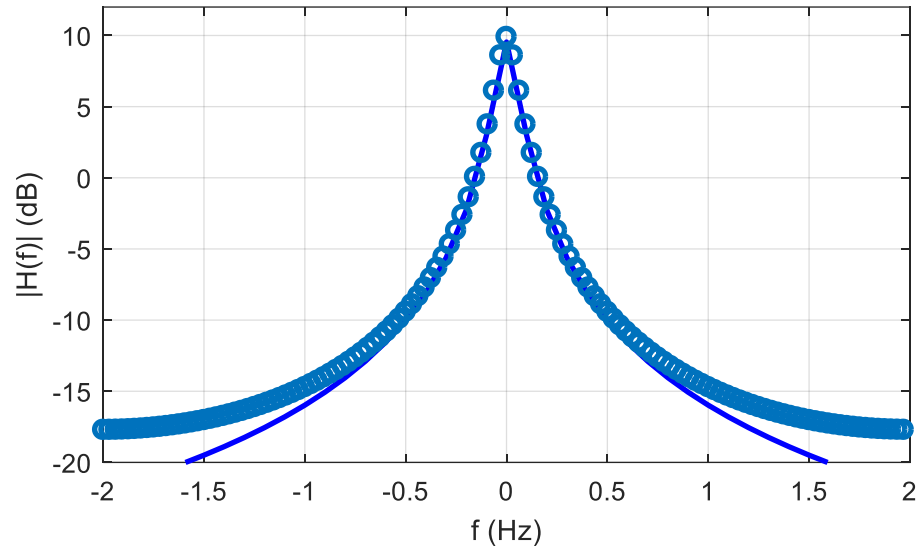
$$|H(f)| = \frac{1}{\sqrt{1 + (2\pi f)^2}}$$

$$\angle H(f) = -\tan^{-1}(2\pi f)$$

$$H[m] = \frac{1 - b^N}{1 - be^{-j2\pi m/N}}$$

$$|H[m]| = \frac{1 - b^N}{\sqrt{1 + b^2 - 2b\cos(2\pi m/N)}}$$

$$\angle H[m] = -\tan^{-1}\left(\frac{\sin(2\pi m/N)}{1 - b\cos(2\pi m/N)}\right)$$



Matched Transfer Function



In the impulse invariance method, the magnitude response of the digital filter did not roll off nearly as fast as the corresponding analog filter. In order to preserve the frequency characteristics, we might try to make the digital system match the analog system in the frequency domain.

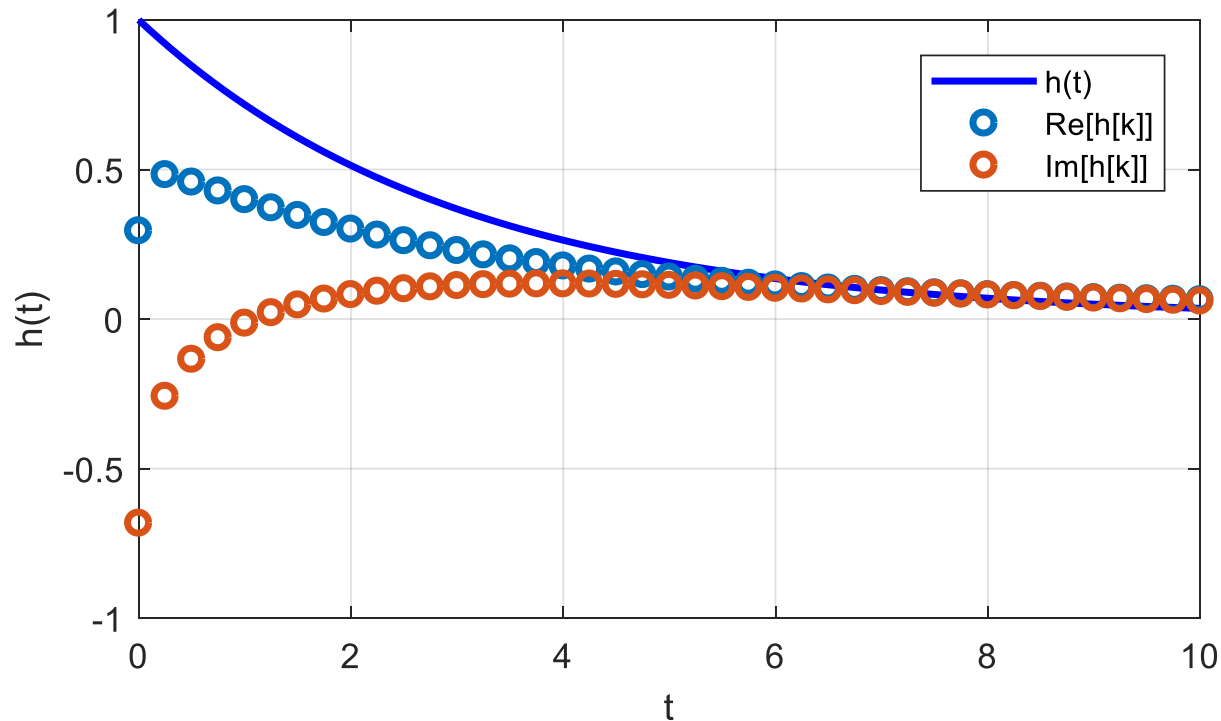
$$H[m] = H(m\Delta f) \leftrightarrow h[k] = \frac{1}{N} \sum_{m=0}^{N-1} H[m] e^{j2\pi km/N}$$

The question is then, how will the digital system impulse response compare with the analog system impulse response?

For the previous example,

$$H[m] = H(m\Delta f) = \frac{1}{a + j2\pi m\Delta f} \leftrightarrow h[k] = \frac{1}{N} \sum_{m=0}^{N-1} \frac{1}{a + j2\pi m\Delta f} e^{j2\pi km/N}$$

Matched Transfer Function



- ✓ When we match the transfer functions, the corresponding impulse response of the DT system is not even necessarily real.
- ✓ We will leave it for a future class (e.g., ECEN 444) to cover how to build DT systems to better match their analog equivalents.